

Table 1. Full MGSM multilingual-transfer results for Qwen3-4B and Llama-3.2-3B-Instruct. Qwen uses 250 problems per language; Llama uses the frozen 100-problem subset. Higher is better for every column.

Model	Language	Method	Accuracy (%) $\uparrow$	Spanish relevance (0–2) $\uparrow$	Instruction relevance (0–2) $\uparrow$	Fluency (0–2) $\uparrow$
Qwen3-4B	Chinese	Original	71.0	0.00	1.80	1.71
		S-PID	0.0	<u>2.00</u>	0.00	0.00
		A-LQR	20.0	1.98	1.53	1.15
		H_∞ (ours)	51.0	2.00	1.87	1.52
	French	Original	60.0	0.00	1.93	1.70
		S-PID	0.0	1.92	0.00	0.00
		A-LQR	29.0	1.94	1.62	1.11
		H_∞ (ours)	48.0	1.96	1.96	1.42
	Japanese	Original	56.0	0.00	1.80	1.64
		S-PID	0.0	<u>2.00</u>	0.00	0.00
		A-LQR	32.0	1.84	1.63	1.01
		H_∞ (ours)	40.0	1.98	1.71	1.20
	Swahili	Original	15.0	0.00	1.43	1.23
		S-PID	0.0	<u>1.96</u>	0.00	0.00
		A-LQR	12.0	1.40	1.32	0.97
		H_∞ (ours)	14.0	0.92	1.44	1.06
	Telugu	Original	6.0	0.00	1.83	1.05
		S-PID	0.0	<u>2.00</u>	0.00	0.00
		A-LQR	<u>19.0</u>	1.88	<u>1.46</u>	<u>1.18</u>
		H_∞ (ours)	8.0	0.34	1.33	1.05
Llama-3.2-3B	Chinese	Original	43.0	0.00	1.79	1.59
		S-PID	9.0	<u>1.96</u>	1.23	0.87
		A-LQR	26.0	1.80	1.49	<u>1.21</u>
		H_∞ (ours)	34.0	0.92	1.50	1.17
	French	Original	49.0	0.00	1.83	1.71
		S-PID	14.0	<u>1.98</u>	1.21	0.67
		A-LQR	<u>43.0</u>	1.24	1.76	1.09
		H_∞ (ours)	42.0	0.80	1.77	1.10
	Japanese	Original	27.0	0.00	1.56	1.34
		S-PID	10.0	<u>1.96</u>	0.67	0.63
		A-LQR	<u>26.0</u>	1.94	<u>1.68</u>	<u>1.31</u>
		H_∞ (ours)	24.0	0.96	1.56	1.28
	Swahili	Original	40.0	0.00	1.79	1.49
		S-PID	2.0	<u>1.90</u>	0.47	0.38
		A-LQR	<u>22.0</u>	0.68	<u>1.47</u>	<u>0.98</u>
		H_∞ (ours)	19.0	0.52	1.38	0.95
	Telugu	Original	6.0	0.00	1.43	1.07
		S-PID	6.0	<u>1.90</u>	0.79	0.46
		A-LQR	<u>21.0</u>	1.86	<u>1.44</u>	<u>1.25</u>